20 In a box, there are only blue sweets, yellow sweets, green sweets and red sweets.
Joan is going to take at random a sweet from the box.

The table shows the probabilities that Joan will take a yellow sweet or will take a red sweet.

Colour	blue	yellow	green	red
Probability		0.3		0.1

The number of green sweets in the box is 3 times the number of blue sweets in the box.

There are 18 yellow sweets in the box.

Work out the number of green sweets in the box.

P(blue or green) = 1 – 0.3 – 0.1 = 0.6

1 mark
All probabilities add to 1.

Blue : green = 1 : 3 → 4 equal parts

P(green) = 0.6 ÷ 4 × 3 = 0.45

1 mark
Green gets 3 of the 4 parts.

18 yellow sweets = 30%, so total = 18 ÷ 0.3 = 60

Green sweets = 0.45 × 60 = 27

2 marks total
Use the total number of sweets to find the green count.

27

(Total for Question 20 is 4 marks)

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